Categorical traces

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Lecture 1. $V \in \mathrm{Vect}_k^{fd}$, $f : V \to V$ - $\mathrm{tr}(f, V) \in k$

$\mathcal{O}$ ambient sym. $\otimes$-cat.

Def $x \in \mathcal{O}$ is dualisable if $\exists\ x^\vee \in \mathcal{O}$, $\left[\begin{array}{l} x \otimes x^\vee \xrightarrow{ev} 1 \\ \cancel{x^\vee \otimes x \to} \\ 1 \xrightarrow{coev} x^\vee \otimes x \end{array}\right.$ s.t.

$$\begin{tikzcd} x \arrow[r, "id \otimes coev"] \arrow[rr, bend right, "id"'] & x \otimes x^\vee \otimes x \arrow[r, "ev \otimes id"] & x \end{tikzcd}$$

$$\begin{tikzcd} x^\vee \arrow[r, "coev \otimes id"] \arrow[rr, bend right, "id"'] & x^\vee \otimes x \otimes x^\vee \arrow[r, "id \otimes ev"] & x^\vee \end{tikzcd}$$

Def $x \in \mathcal{O}^{dual}$, $f : x \to x$, $\mathrm{tr}(f, x) \in \mathrm{End}_{\mathcal{O}}(1)$

$$1 \xrightarrow{coev} x \otimes x^\vee \xrightarrow{f \otimes id} x \otimes x^\vee \xrightarrow{ev} 1$$

Main situation. $\mathcal{O} = \mathrm{Pr}_k^L$ = $(\infty, 1)$ cat of presentable stable k-linear cats

$\mathrm{Pr}^{dualisable}$ difficult to describe.

Lem $\mathcal{C} \in \mathrm{Pr}$ compactly generated, then $\mathcal{C}$ is dualisable.

$\mathcal{C} = \mathrm{Ind}\, \underbrace{\mathcal{C}_0}_{\text{cat of cpt obj.}}$, $\mathcal{C}^\vee = \mathrm{Ind}\, \mathcal{C}_0^{op}$

$\mathcal{C} \otimes \mathcal{C}^\vee \to D(k)$ ind completion of $\mathcal{C}_0 \times \mathcal{C}_0^{op} \to D(k)$

$(A, B) \mapsto \mathrm{Maps}(B, A)$

Eg. $C = D(R)$, R k-Alg

$= \mathrm{Ind\,Perf}(R)$

$R^{rev} = R$ as a k-vec.sp. $x^{rev} \cdot y^{rev} = yx$

$$(\mathrm{Perf}\, R)^{op} \xrightarrow{\sim} \underbrace{\mathrm{Perf}\ R^{rev}}_{\text{Perf (right } R\text{-mod)}}$$

$$M \longmapsto \mathrm{Hom}(M, R)$$

$$\text{eval}: \ D(R) \otimes D(R^{rev}) \longrightarrow D(k)$$

$$\begin{tikzcd} D(R) \otimes D(R^{rev}) \arrow[d, equal] \arrow[r] & D(k) \\ D(R \otimes R^{rev}) \arrow[ur] & \end{tikzcd}$$

R^{rev} viewed as an $(R \otimes R^{rev}, k)$-bimod.

$$\text{coev}: \ D(k) \longrightarrow D(R) \otimes D(R^{rev}) \simeq D(R \otimes R^{rev})$$

$$\begin{tikzcd} D(k) \arrow[rr, bend right, "R"'] \arrow[r] & D(R) \otimes D(R^{rev}) & D(R \otimes R^{rev}) \end{tikzcd}$$

R viewed as a bimodule

$$\mathrm{tr}(\mathrm{id}, D(R)) \qquad \begin{tikzcd} D(k) \arrow[r, "\text{coev}", "R"'] \arrow[rr, bend right] & D(R) \otimes D(R^{rev}) \arrow[r, "\text{ev}", "R^{rev}"'] & D(k) \end{tikzcd}$$

$$R \otimes_{R \otimes R^{rev}} R^{rev} = HH(R)$$

Hochschild homology of R

$D(R) \xrightarrow[M]{} D(R)$, M some R bimodule

$$\mathrm{tr}(M, D(R)): \quad D(k) \xrightarrow[HH(R,M)]{} D(k)$$

$$R^{rev} \otimes_{R \otimes R^{rev}} (M \otimes R) \otimes_{R \otimes R^{rev}} R \simeq M \otimes_{R \otimes R^{rev}} R$$

Interlude. Bar complex

$$x \otimes y \longmapsto xy$$

$$\cdots \ \substack{\rightarrow \\ \rightarrow \\ \rightarrow} \ R \otimes R \otimes R \rightrightarrows R \otimes R \longrightarrow R$$

$$x \otimes y \otimes z \mapsto xy \otimes z$$

$$\mapsto x \otimes yz$$

gives a resolution of R as R-bimod $= R \otimes R^{rev}$-mod

Usual theory of HH, for a bimodule M

$$HH(R,M) = \left| \cdots R\otimes R\otimes M \substack{\to\\ \to\\ \to} R\otimes M \rightrightarrows M \right|$$

$$x\otimes m \mapsto xm$$
$$\mapsto mx$$

$$x\otimes y\otimes m \mapsto xy\otimes m$$
$$\mapsto x\otimes ym$$
$$\mapsto y\otimes m\cancel{x}$$

<u>Functorial properties</u>

(i) transposition $\quad C\in Pr^{dual} \quad f: C\to C$

$$C^\vee \xrightarrow{f^\vee} C^\vee$$

$$tr(f, C) \simeq tr(f^\vee, C^\vee)$$

(ii) cyclicity $\quad C\xrightarrow{f} D\xrightarrow{g} C \quad C, D$ dualizable

$$tr(f\circ g, D) \simeq tr(g\circ f, C)$$

(iii) monoidality $\quad C\xrightarrow{f} C \quad C, D$ dualisable, $C\otimes D$ again dualisable

$D\xrightarrow{g} D$

$$tr(f\otimes g, C\otimes D) \simeq tr(f, C)\otimes tr(g, D)$$

(iv) functoriality $\quad C\xrightarrow{f} C, \; D\xrightarrow{g} D \quad$ dualisable

$\phi: C\to D$ which admits a right adjoint

ϕ^R exists as a functor, we assume that it lies in Pr meaning

ϕ^R commutes with colimits.

$$\begin{tikzcd} C \arrow[r, "\phi"] \arrow[d, "f"'] & D \arrow[d, "g"] \\ C \arrow[r, "\phi"'] \arrow[ur, phantom, "\nearrow\alpha"] & D \end{tikzcd}$$

$\alpha : \phi f \longrightarrow g \phi$

$\mathrm{tr}(\phi, \alpha) : \mathrm{tr}(f, C) \to \mathrm{tr}(g, D)$

is the following composition

$$\mathrm{tr}(f, C) \xrightarrow{\text{unit } \phi} \mathrm{tr}(\phi^R \phi f, C) \xrightarrow{\text{cycl}} \mathrm{tr}(\phi f \phi^R, D)$$

$$\xrightarrow{\alpha} \mathrm{tr}(g \phi \phi^R, D) \xrightarrow{\text{counit } \phi} \mathrm{tr}(g, D)$$

Chern characters $\quad C \in \mathrm{Pr}^{\text{dualizable}}$, $f : C \to C$

$$(D(k), \mathrm{id}) \overset{(\phi, \alpha)}{\rightsquigarrow} (C, f)$$

$\phi : D(k) \to C$ is the datum of $C_\phi \in C$

$\quad k \mapsto C_\phi$

ϕ^R continuous $\iff$ C_ϕ compact.

$\alpha : C_\phi \longrightarrow f(C_\phi)$

$$\begin{tikzcd} \mathrm{tr}(\mathrm{id}, D(k)) \arrow[r, "\mathrm{tr}(\phi, \alpha)"] \arrow[d, phantom, "\parallel"] & \mathrm{tr}(C, f) \arrow[d, phantom, "\rotatebox{90}{$\ni$}"] \\ k \arrow[ur] \arrow[r] & \mathrm{cl}(C_\phi, \alpha) \in H^0(\mathrm{tr}(C, f)) \\ 1 \arrow[ur, mapsto] & \end{tikzcd}$$

Example $\quad C = D(\mathrm{Rep}_k \Gamma)$, Γ finite gp. char $k = 0$

$f = \mathrm{id}$ acting on C; $D(\mathrm{Rep}_k \Gamma) = D(k\Gamma\text{-mod})$

$$\mathrm{tr}(\mathrm{id}, D(\mathrm{Rep}_k \Gamma)) = Z(k[\Gamma]) = \text{"space of characters of rep of } \Gamma\text{"}$$

$M \in \mathrm{Rep}^{\text{f.d.}}_k \Gamma \rightsquigarrow M \in D(\mathrm{Rep}_k \Gamma)$ is compact, $\alpha : M \xrightarrow{\mathrm{id}} M$

$\mathrm{cl}(M, \alpha) \in Z(k\Gamma)$ is the usual char of M.

<u>Eg.</u> Grothendieck sheaf func.-dict.

X f.t. scheme $/\mathbb{F}_q$, $X = X_0 \underset{\mathbb{F}_q}{\times} \overline{\mathbb{F}_q} \curvearrowleft F$ geom. Frob.

$C = \mathrm{Shv}(X, \overline{\mathbb{Q}_\ell}) \curvearrowleft F_*$

$\mathcal{F} \in \underset{\substack{\cup \\ \text{constructible}}}{\mathrm{Shv}_c}(X, \overline{\mathbb{Q}_\ell})$ with a Weil structure $\mathcal{F} \xrightarrow{\alpha} F_* \mathcal{F}$

$\mathrm{tr}(\alpha, \mathcal{F}) : X_0(\mathbb{F}_q) \longrightarrow \overline{\mathbb{Q}_\ell}$

$x \longmapsto \mathrm{tr}(\alpha_x, \mathcal{F}_x)$

$x^* \mathcal{F} \xrightarrow{\alpha_x} x^* F_* \mathcal{F} \simeq x^* \mathcal{F}$

Ideally, $\mathrm{tr}(F_*, \mathrm{Shv}(X, \overline{\mathbb{Q}_\ell})) \overset{?}{\not\simeq} \mathrm{Fun}(X_0(\mathbb{F}_q), \overline{\mathbb{Q}_\ell})$

$\mathrm{Shv}(X, \overline{\mathbb{Q}_\ell}) = \mathrm{Shv}(X) = \mathrm{Ind}\, \mathrm{Shv}_c(X, \overline{\mathbb{Q}_\ell})$

$\mathbb{D} : \mathrm{Shv}_c(X)^{op} \xrightarrow{\sim} \mathrm{Shv}_c(X)$ induces an iso. $\mathrm{Shv}(X)^\vee \xrightarrow{\sim} \mathrm{Shv}(X)$

<u>Lem.</u>
$$\left.\begin{aligned} \mathrm{Shv}(X) \otimes \mathrm{Shv}(X) &\to D(\overline{\mathbb{Q}_\ell}) \\ (A \otimes B) &\mapsto R\Gamma(X, \Delta^!(A \boxtimes B)) \end{aligned}\right] \text{ev.}$$

$$\mathrm{Shv}(X) \otimes \mathrm{Shv}(X) \underset{\boxtimes^R}{\overset{\boxtimes}{\rightleftarrows}} \mathrm{Shv}(X \times X)$$

<u>coev:</u>
$$\begin{aligned} D(\overline{\mathbb{Q}_\ell}) &\longrightarrow \mathrm{Shv}(X) \otimes \mathrm{Shv}(X) \\ \overline{\mathbb{Q}_\ell} &\longmapsto \boxtimes^R \Delta_* \omega_X \leftarrow \text{dualizing sheaf} \end{aligned}$$

Construction of GV:

$$\begin{tikzcd}
 & \mathrm{Shv}(X) \otimes \mathrm{Shv}(X) \arrow[r, "F_* \otimes id"] \arrow[d, "\boxtimes"] \arrow[dr, phantom, "\Downarrow \wr"] & \mathrm{Shv}(X) \otimes \mathrm{Shv}(X) \arrow[d, "\boxtimes"] \arrow[dr] \arrow[dr, phantom, "\Downarrow \wr", near start] & \\
D(\overline{\mathbb{Q}_\ell}) \arrow[ur, "coev"] \arrow[r, "\Delta_* \omega_X"'] \arrow[ur, phantom, "\Downarrow \odot", near end] \arrow[rrr, bend right=40, "R\Gamma(X^F, \omega_X) = \mathrm{Fun}(X^F, \overline{\mathbb{Q}_\ell})"'] & \mathrm{Shv}(X \times X) \arrow[r, "(F \times id)_*"'] & \mathrm{Shv}(X \times X) \arrow[r, "R\Gamma(X, \Delta^! -)"'] & D(\overline{\mathbb{Q}_\ell})
\end{tikzcd}$$

($*$) $\boxtimes \circ \mathrm{coev} = \boxtimes\, \boxtimes^R\, \Delta_* \omega_X \longrightarrow \Delta_* \omega_X$

Defines $\mathrm{tr}(F_*, \mathrm{Shv}(X)) \longrightarrow \mathrm{Fun}(X^F, \overline{\mathbb{Q}_\ell})$

Th given $(\mathcal{F}, \alpha)$ a Weil struct, $\mathrm{cl}(\mathcal{F}, \alpha) \in H^0(\mathrm{tr}(F_*, \mathrm{Shv}(X)))$

$$\downarrow$$

is the function of the dictionary $\mathrm{Fun}(X^F, \overline{\mathbb{Q}_\ell})$

Lecture 2 $\quad$ $\mathcal{O}$ ambient monoidal cat. $2\text{-}\mathrm{Pr}$

In practice, all examples that we care about are of the form $A\text{-mod}(\mathrm{Pr})$

for $A \in \mathrm{Alg}(\mathrm{Pr})$

e.g. R comm. k-Alg $(D(R), \otimes)$, $\begin{matrix} X \\ \downarrow \\ \mathrm{Spec}\, R \end{matrix}$ $\quad$ $\mathrm{QCoh}(X) \in D(R)\text{-Mod}$

$A \in \mathbf{Alg}(\mathrm{Pr})$, A-mod, $f: A \to A$ $\otimes$-endofunctor

$$A\text{-mod} \xrightarrow[=:f]{\mathrm{Res}_f} A\text{-mod} \quad \text{given by an } A\text{-module} \quad \underline{A_f}$$

right module structure twisted by f

$$\mathrm{Tr}(f, A\text{-mod}) = HH(A, A_f) = A \underset{A \otimes A^{\mathrm{rev}}}{\otimes} A_f$$

[duality datum on A-mod $\quad$ $\mathbb{1} \in 2\text{-}\mathrm{Pr}$ is Vect_k-mod

$$\mathrm{Vect}_k\text{-mod} \xrightarrow{A} A\text{-mod} \otimes A^{\mathrm{rev}}\text{-mod} = \underbrace{(A \otimes A^{\mathrm{rev}})}_{\in \mathrm{Alg}(\mathrm{Pr})}\text{-mod}$$

$$A^{\mathrm{rev}}\text{-mod} \otimes A\text{-mod} \xrightarrow{A^{\mathrm{rev}}} \mathrm{Vect}_k\text{-mod}$$

Ex. $\begin{matrix} X \\ \downarrow \\ \mathrm{Spec}\, k \end{matrix}$ scheme f.t. $\qquad f: X \to X$ some endomorphism

$\qquad f^*: \mathrm{QCoh}(X) \to \mathrm{QCoh}(X)$ $\otimes$-endomorphism

$$\mathrm{Tr}(f^*, \mathrm{QCoh}(X)\text{-mod}) = \mathrm{QCoh}(X) \underset{\mathrm{QCoh}(X)\otimes \mathrm{QCoh}(X)}{\otimes} \mathrm{QCoh}(X)_f$$

$$= \mathrm{QCoh}\left(X \underset{\Delta \to X\times X \leftarrow \mathrm{id}\times f}{\times} X\right)$$

rel. Künneth formula

$$X \underset{X\times X}{\times} X = L_f X \quad f\text{-loop stack of } X$$

$$= X^f \quad \text{derived } f\text{-fixed points}$$

$$= f\text{-inertia stack}$$

ex $f = \mathrm{id} \rightsquigarrow$ intersection is not transverse.

ex $f = \mathrm{id}$, $X = \mathrm{Spec}\, R$. $\quad LX = \mathrm{Spec}(R \underset{R\otimes R}{\otimes} R) = \mathrm{Spec}(HH(R,R))$

extension to stacks over k of char 0, Ben-Zvi-Francis-Nadler

$\rightsquigarrow$ large class of stacks for which these Künneth formulae hold

includes any quotient X/G where G reductive.

eg $f: G \to G$. $\quad L_f(*/G) \simeq G/\mathrm{Ad}_f G$, f-twisted conj. $g.x = g x f(g^{-1})$.

Deligne-Lusztig theory

$G/\overline{\mathbb{F}_q} \circlearrowleft F$ geom. Frobenius coming from some $\mathbb{F}_q$-structure.

$\mathrm{Shv}(G)$ = cat. of ℓ-adic sheaves on G

monoidal cat. for convolution

$A, B \in \mathrm{Shv}(G)$, $\quad A * B = m_!(A \boxtimes B)$, $\quad m: G\times G \to G$

In an ideal world, $f^*: \mathrm{Shv}(G) \to \mathrm{Shv}(G)$, $\mathrm{Tr}(f^*, \mathrm{Shv}(G)\text{-mod})$

$$= \mathrm{Shv}(G) \underset{\mathrm{Shv}(G)\otimes \mathrm{Shv}(G)}{\otimes} \mathrm{Shv}(G)_F$$

ideally want $\quad \mathrm{Shv}\left(\Delta_G \backslash G \times G / \Delta_F G\right) = \mathrm{Shv}\left(\frac{G}{\mathrm{Ad}_F G}\right)$

$$\Delta_F : G \to G \times G$$
$$g \mapsto (g, Fg)$$

G conn'd, Lang's th $\quad G/\mathrm{Ad}_F G \simeq */G^F. \quad G^F = G(\mathbb{F}_q)$

Several solutions to remedy Künneth formula

- Gaitsgory - Rozenblyum - Varshavsky (uses AGCat formalism)

- replace $\mathrm{Shv}(G)$ by something Morita equivalent

if instead of ℓ-adic sheaves, use D-modules, then Künneth formulae are true

Th [BZ - GN - O] In the D-module setting,

$$\mathrm{Shv}(G)\text{-mod} \xrightarrow{\sim} \mathrm{Shv}(U\backslash G/U)\text{-mod}$$
$$\mathrm{Shv}(G/U)$$

where $U \subset B \subset G$ Borel subgp.

Back in the ℓ-adic setting, $\mathrm{Shv}(U\backslash G/U)$ already too big.

Solution: pass to the cat. of monodromic sheaves. $\quad \mathrm{Shv}(\mathbb{A}^1) \otimes \mathrm{Shv}(\mathbb{A}^1) \xrightarrow[\times]{} \mathrm{Shv}(\mathbb{A}^2)$

Interludes: monodromic sheaves.

T torus, $\quad \mathrm{Shv}(T) \supset \mathrm{Shv}(T)_{mon} = \left\{ A \in \mathrm{Shv}(T) : \forall i, H^i(A) \text{ is a tame local systems} \right\}$

$$\pi_1^{\acute{e}t}(T, 1) \to \pi_1^t(T, 1) = \varprojlim_{(n,p)=1} T[n] \simeq X_*(T) \otimes \hat{\mathbb{Z}}^{(p)}(1)$$

$\varprojlim_n T \to T$ $\quad$ transitions are $T \to T$, $t \mapsto t^n$

$\mathrm{Shv}_c(T)_{mon} = \langle \mathcal{L}_\chi : \chi : \pi_1^t(T) \to \overline{\mathbb{Q}}_\ell^\times$ of finite order $\rangle$

$\cup$

$\mathrm{Shv}_c(T)_{\chi\text{-mon}} = \langle \mathcal{L}_\chi \rangle = \mathrm{Coh}((1 \subset \hat{T})^\wedge)$

$\hat{T}$ torus dual to $T / \overline{\mathbb{Q}}_\ell$. $1 \in \hat{T}$

$\mathrm{Shv}(T)^\wedge_{\chi\text{-mon}} \simeq \mathrm{QCoh}(\mathrm{Spec}(\mathcal{O}(\hat{T})^\wedge_1))$ free monodromic sheaves due to Z. Yun

monoidal under $*$ $\longleftrightarrow$ $\otimes$

$\mathcal{L} \longleftrightarrow \mathcal{O}$

$\mathcal{H} = \mathrm{Shv}(U\backslash G/U)^\wedge_{mon} \supset \mathrm{Shv}_c(U\backslash G/U)_{mon} = \{ A \in \mathrm{Shv}_c(U\backslash G/B)$ s.t. pullback to any ~~T~~ orbit is in $\mathrm{Shv}_c(T)_{mon} \}$

Th $\begin{pmatrix} \text{E. '25} \\ \text{Zhu '25} \\ \text{G-R-V '26} \end{pmatrix}$ $\quad \mathrm{Tr}(F^*, \mathcal{H}\text{-mod}) \simeq \mathrm{Shv}(G/\mathrm{Ad}_F G)$

$\mathcal{H} \otimes \mathcal{H}_F$

$\mathcal{H} \otimes \mathcal{H}^{rev}$

$\mathcal{H}_T \hookrightarrow \mathcal{H}$

$\shortparallel$

$\mathrm{Shv}(T)^\wedge_{mon}$

write $\mathcal{H} = | \mathcal{H} \otimes_{\mathcal{H}_T} \mathcal{H} \leftleftarrows \mathcal{H} \otimes_{\mathcal{H}_T} \mathcal{H} \otimes_{\mathcal{H}_T} \mathcal{H} \leftleftarrows \cdots |$

$\wr$

$\mathcal{H} \otimes_{\mathcal{H}\otimes\mathcal{H}^{rev}} \mathcal{H}_F = | \underbrace{(\mathcal{H} \otimes_{\mathcal{H}_T} \mathcal{H}) \otimes_{\mathcal{H}\otimes\mathcal{H}^{rev}} \mathcal{H}_F}_{\mathcal{H}_T \otimes_{\mathcal{H}_T\otimes\mathcal{H}_T} \mathcal{H}_F} \leftleftarrows \cdots |$

$\leftleftarrows \cdots$

Parch

$$= \left| \mathrm{Shv}\left(\frac{U\backslash G/U}{\mathrm{Ad}_F T}\right) \leftleftarrows \mathrm{Shv}\left(\frac{U\backslash G/U \overset{T}{\times} U\backslash G/U}{\mathrm{Ad}_F T}\right) \begin{smallmatrix}\leftarrow\\ \leftarrow\\ \leftarrow\end{smallmatrix} \cdots \right|$$

$$\frac{G}{\mathrm{Ad}_F B}$$

$$\downarrow$$

$$\frac{G/U}{\mathrm{Ad}_F B} = \frac{U\backslash G/U}{\mathrm{Ad}_F T} \quad \underset{U\text{-equiv.}}{\xleftrightarrow{\text{forget}}} \quad \underbrace{\left| \mathrm{Shv}\left(\frac{G}{\mathrm{Ad}_F B}\right) \leftleftarrows \mathrm{Shv}\left(\frac{G \overset{B}{\times} G}{\mathrm{Ad}_F B}\right) \begin{smallmatrix}\leftarrow\\ \leftarrow\\ \leftarrow\end{smallmatrix} \cdots \right|}$$

nerve of the map $\frac{G}{\mathrm{Ad}_F B} \to \frac{G}{\mathrm{Ad}_F G}$

$$\simeq \mathrm{Shv}\left(\frac{G}{\mathrm{Ad}_F G}\right)$$

$$\begin{array}{ccc} \mathrm{Tr}(F^*, \mathcal{H}\text{-mod}) & \hookrightarrow & \mathrm{Shv}\left(\frac{G}{\mathrm{Ad}_F G}\right) \\ \uparrow & & \\ \mathcal{H} & & \end{array} \qquad \begin{array}{c} 1\otimes x \\ \uparrow \\ x \end{array}$$

$\rightsquigarrow$ find some nice objects $\Delta_w \in \mathcal{H}$

$$ch(\Delta_w) = R\Gamma_c(Y(w), \overline{\mathbb{Q}}_\ell^{\times})$$

Th (Deligne - Lusztig '76) All irreps of G^F appears in the cohomology of DL-var.

$\Rightarrow$ essential surjectivity.

Lecture 3. $G/\mathbb{F}_q \circlearrowleft F$, $B = TU$ stable under F

$\mathcal{H}$ = "free monodromic" finite Hecke category $= \mathrm{Shv}(U\backslash G/U)^{\wedge}_{mon}$

Th. $\mathrm{Tr}(\mathrm{Frob}^*, \mathcal{H}\text{-mod}) \simeq \mathrm{Shv}(G/\mathrm{Ad}_F G) = D(\mathrm{Rep}_\Lambda G^F)$ $\quad = \mathrm{Shv}(G/\mathrm{Ad}_F G)$

$$\underset{\mathcal{H}\otimes\mathcal{H}^{rev}}{\mathcal{H} \overset{\shortparallel}{\otimes} \mathcal{H}_F} = \left| \mathrm{Shv}\left(\frac{U\backslash G/U}{\mathrm{Ad}_F T}\right) \leftleftarrows \mathrm{Shv}\left(\frac{U\backslash G/U \overset{T}{\times} U\backslash G/U}{\mathrm{Ad}_F T}\right) \begin{smallmatrix}\leftarrow\\ \leftarrow\\ \leftarrow\end{smallmatrix} \cdots \right)$$

$$\begin{tikzcd} \mathcal{H} \otimes \mathcal{H}_F \\ H \otimes H^{\mathrm{rev}} \\ H \arrow[u, "t"] \end{tikzcd}$$

$$\begin{tikzcd} U \backslash G / U & G/\mathrm{Ad}_F U \arrow[l] \arrow[d] \arrow[drr] & & \\ \dfrac{U\backslash G/U}{\mathrm{Ad}_F T} \arrow[from=u, "\pi"] & G/\mathrm{Ad}_F B \arrow[l, "\alpha"] \arrow[rr, "\beta"] & & \dfrac{G}{\mathrm{Ad}_F G} \end{tikzcd}$$

Lem There is an. isom. of functors $\mathcal{H} \to \mathrm{Shv}(G/\mathrm{Ad}_F G)$

$$t \simeq \beta_! \alpha^* \pi_!$$

Recall "Classical DL theory": $w \in W$, $\dot{w} \in N(T)$ lift of w

$$\begin{tikzcd} G^F \curvearrowright Y(\dot{w}) \arrow[d] = \{gU : g^{-1}Fg \in U\dot{w}U\} \subset G/U \arrow[d] \curvearrowleft T^{wF} \\ G^F \curvearrowright X(w) = \{gB : (g^{-1}Fg) \in BwB\} \subset G/B \end{tikzcd}$$

$$R\Gamma_c(Y(\dot{w}), \overline{\mathbb{Q}}_\ell)$$

Rem. $\dfrac{G}{\mathrm{Ad}_F B} \xleftarrow{\sim} G^F \backslash G/B$

$$g^{-1}Fg \longleftarrow\!\shortmid\, g$$

$$\begin{tikzcd} Y/T^{wF} \ltimes (U^w \cap U) \left[\dfrac{U\backslash BwB/U}{\mathrm{Ad}_F T} \right. & G^F \backslash Y(\dot{w}) / T^{wF} \arrow[l] \arrow[d] \arrow[dr] & \\ \dfrac{U\backslash G/U}{\mathrm{Ad}_F T} \arrow[u, phantom, "\cap"] & \dfrac{G}{\mathrm{Ad}_F B} \arrow[l] \arrow[r] & G/\mathrm{Ad}_F G = */G^F \end{tikzcd}$$

$$\begin{array}{ccc}
Shv(U\backslash BwB/U)^{\wedge}_{mon} & \underset{j_{w!}}{\subset} & Shv(U\backslash G/U)^{\wedge}_{mon} \\
\wr\!\! | & & \cup \\
Shv(T\cdot \dot{w})^{\wedge}_{mon} & & \mathcal{L}_w \\
\wr\!\! | & & \uparrow \\
QCoh(\coprod_{\chi \text{ torsion}} \hat{T}^{\wedge}_{\chi}) & & \mathcal{O}
\end{array}$$

p^{+}

$$\Delta_w = j_{w!}\, \mathcal{L}_w [\ell(w) + \dim T]$$

$$\pi_! \Delta_w \in Shv\left(\frac{U\backslash G/U}{Ad_F T}\right)$$

$$= j_{w!}(Reg_{T^{wF}})$$

$$\in Shv\left(\frac{U\backslash BwB/U}{Ad_F T}\right)$$

$$\simeq Shv(*/T^{wF} \times (U\cap U^w))$$

Parabolic version

Classical Deligne–Lusztig construction

$$B \subset P \subset G$$
$$\cup \quad\quad \cup$$
$$T \subset L$$

$v \in W$ s.t. $\quad Ad(v)\circ F(L) = L$

$$G^F \circlearrowright Y(P,\dot{v}) = \{ g \in G/U_P : g^{-1}Fg \in U_P \dot{v} F U_P \} \circlearrowleft L^{\dot{v}F}$$

$$H_G \xleftarrow{i_P} H_L$$

$$Shv(U\backslash G/U)^{\wedge}_{mon} \xleftarrow[i_P]{i_* q^*} Shv(U_L\backslash L/U_L)^{\wedge}_{mon}$$

$$U\backslash G/U \xleftarrow{i} U\backslash P/U \xrightarrow{q} U_L\backslash L/U_L$$

i_P is monoidal

$$\mathcal{H}^{P,v} = Shv(U\backslash PvFP/U)^{\wedge}_{mon} \subset \mathcal{H}$$

stable under H_L

$\rightsquigarrow \mathcal{H}_F^{P,\nu}$ is an $\mathcal{H}_L$-bimodule

Can consider $\mathrm{Tr}(\mathcal{H}_F^{P,\nu}, \mathcal{H}_L\text{-mod})$

Th. (i) $\simeq \mathrm{Shv}\left(\dfrac{U_P\backslash P\nu F P/U_P}{\mathrm{Ad}_F L}\right)$

$$= */L^{\nu F} \ltimes (\underbrace{U_P \cap {}^{\nu}F(U_P)}_{\text{conn'd unip group}})$$

inclusion of $\mathcal{H}^{P,\nu} \hookrightarrow \mathcal{H}_G$

$$(\mathcal{H}_F^{P,\nu}, \mathcal{H}_L\text{-mod}) \xrightarrow{j_{\nu !}\mathcal{H}_G} (\mathcal{H}_{G,F}, \mathcal{H}_G\text{-mod})$$

viewed as an $(\mathcal{H}_G, \mathcal{H}_L)$-mod

(ii) $\mathrm{Tr}(j_{\nu !}, \mathcal{H}_G) : \mathrm{Tr}(\mathcal{H}_F^{P,\nu}, \mathcal{H}_L\text{-mod}) \to \mathrm{Tr}(\mathcal{H}_{G,F}, \mathcal{H}_G\text{-mod})$ is the functor

$$\begin{tikzcd}
\dfrac{U_P\backslash P\nu F P/F(U_P)}{\mathrm{Ad}_F L} \arrow[r, hook, "j_\nu"] & \dfrac{U_P\backslash G/FU_P}{\mathrm{Ad}_F L} & \\
U^P\backslash Y(\bar{\nu}, P)/L^{\nu F} \arrow[u] \arrow[r] & \dfrac{G}{\mathrm{Ad}_F P} \arrow[u, "\alpha_P"] \arrow[r, "\beta_P"] & \dfrac{G}{\mathrm{Ad}_F G}
\end{tikzcd}$$

$\leftarrow$ parabolic character sheaves

$\beta_{P!}\, \alpha_P^* \, j_{\nu !}$

(iii) $\mathrm{Tr}(\mathcal{H}_{G,F}, \mathcal{H}_L\text{-mod}) = \mathrm{Shv}\left(\dfrac{U_P\backslash G/FU_P}{\mathrm{Ad}_F L}\right)$

1-cat. traces & 2-cat traces

A mon. cat. $\quad f : A \xrightarrow{\otimes} A \qquad A \rightsquigarrow \mathrm{tr}(f, A) \in \mathrm{Vect}_k$

$f \circlearrowright A\text{-mod} \rightsquigarrow \mathrm{Tr}(f, A\text{-mod}) \in \mathrm{Pr}$

Recall $\mathrm{tr}\left(f_1 \otimes f_2, A \otimes B\right) = \mathrm{tr}(f_1, A) \otimes \mathrm{tr}(f_2, B)$

$\Rightarrow$ trace sends an algebra to an algebra

$\mathrm{tr}(f, A) \in \mathrm{Alg}(\mathrm{Vect}_k) \rightsquigarrow \mathrm{tr}(f, A)\text{-mod}$

<u>Def</u> $\mathcal{A}$ monoidal compactly gen. cat.

$\mathcal{A}$ is rigid if

(i) every compact object is left and right dualisable

(ii) $\mathcal{A} \otimes \mathcal{A} \xrightarrow{*} \mathcal{A}$ is continuous with continuous right adj. ($\Leftrightarrow$ $*$ preserves cpt obj')

(iii) $1 \in \mathcal{A}$ compact

<u>ex</u> – X qc scheme. $\mathrm{QCoh}(X)$ is rigid

– $\mathcal{H}$ the "free monodromic" Hecke cat.

$\mathcal{A}$ a rigid mon. cat. $\circlearrowright f$

$$\mathrm{Tr}(f, \mathcal{A}\text{-Mod}) \longleftarrow \mathcal{A}$$
$$\parallel$$
$$\mathcal{A} \otimes_{\mathcal{A} \otimes \mathcal{A}^{rev}} \mathcal{A}_f \qquad 1 \otimes x \longleftarrow\!\shortmid x$$

Description of this map in terms of Chern character.

<u>Data</u>. $\mathcal{M} \in \mathcal{A}\text{-mod}$ (right dualisable as $\mathcal{A}$-mod $\Leftrightarrow$ $\mathcal{M}$ is $\mathcal{A}$-rigid dualisable as a plain category)

$\mathcal{M} \xrightarrow{\alpha} f^*(\mathcal{M})$

$\rightsquigarrow \mathrm{cl}(\mathcal{M}, \alpha) \in \mathrm{Tr}(f, \mathcal{A}\text{-mod})$

$x \in \mathcal{A}$, define $\mathcal{M} = \mathcal{A}$ viewed as a left $\mathcal{A}$-mod, $\alpha : \mathcal{A} \longrightarrow \mathcal{A}$, $y \mapsto f(y) \cdot x$

Recall

Lem There is a canonical isom. in $\mathrm{Tr}(f, A\text{-mod})$

$$t(x) \simeq cl(A, \alpha)$$

$$\begin{array}{c} A \\ \updownarrow t \\ \mathrm{Tr}(f, A\text{-mod}) \end{array}$$

Th [GKRV] $f \circlearrowright A$ rigid mon. cat.

$M \in A\text{-mod}$, M dualisable, $\alpha : M \to f(M)$

(i) $t(1) \in \mathrm{Tr}(f, A\text{-mod})$

There exists an iso. of algebras $\mathrm{End}(t(1)) \simeq tr(f, A)$

(ii) $\mathrm{Hom}(t(1), cl(M, \alpha)) \simeq tr(\alpha, M)$

Ex. (i) take $A = \mathrm{QCoh}(X)$, X f.t. scheme $/k$

- $\mathrm{Tr}(id, A\text{-Mod}) \simeq \mathrm{QCoh}(\mathcal{L}X)$
- $tr(id, \mathrm{QCoh}(X)) \simeq HH(X) = R\Gamma(\mathcal{L}X, \mathcal{O})$

(ii) take $\mathcal{H}$ to be the free mon. finite Hecke cat.

$$\begin{array}{ccc} \mathrm{Tr}(\mathrm{Frob}, \mathcal{H}\text{-mod}) & \simeq & \mathrm{Shv}(*/G^F) \\ \uparrow & & \Downarrow \\ 1 \in \mathcal{H} & \longmapsto & \mathrm{ind}_{B^F}^{G^F} 1 \end{array}$$

$$\mathrm{End}(\mathrm{ind}_{B^F}^{G^F} \mathbb{1}) = \text{Hecke algebra} = tr(\mathrm{Frob}, \mathcal{H})$$

function sheaf dictionary

and local term of lecture 1.

further application to Deligne–Lusztig theory.

- endoscopy for (modular) rep'n of finite gps of Lie type — Lusztig–Yun, '20 / char; E. '26 / modl
- independence of parabolic for DL induction